

Question D.1 — Joint Instrumentation of Repair, Error and Detector

All measurements come from the instrumented campaign of 2 000 runs (20 amplitudes $\times$ 100 seeds, $H = 2000$, $M = 10$ trees in the adaptive random forest, ARF) in `04_experimentations/resultats_R2/`. Every figure quoted is reproduced by `scripts/chiffres_QD.py` and `scripts/derives_QD.py`. The hierarchy $\tau_{\text{ARF}} \leq \tau_{\text{swap}}(q)$ of Q C.1 is assumed established.

1 What Is Recorded and What Is Recomputed

The simulation writes raw material only: the error trajectory $e_t \in \{0, 1\}$, the tree replacement events, and the 3 000 pre-drift steps. No threshold and no λ appears anywhere in the simulation loop.

That is deliberate. The external detector reads the error trajectory and never acts on the forest, so λ has no influence on the dynamics. Once the **break** is removed, the three scenarios of R2 produce runs identical to the bit. One campaign replaces three, and S_t , τ_{det} , $A(H)$ and $S_{\text{max}}(H)$ are swept offline for every threshold and any δ_P . Freezing a threshold inside the loop would have cost one campaign per threshold, and at $\lambda = 50$ it would have produced a table whose τ_{det} column is censored in 99.95% of runs.

Notation. τ^* is the drift instant; every duration below is counted from it, and the detector is created at τ^* with $S_{\tau^*} = 0$. τ_{ARF} is the first tree replacement after τ^* ; $\tau_{\text{swap}}(q)$ the first instant at which a fraction q of the M trees has been replaced, so that $\tau_{\text{ARF}} = \tau_{\text{swap}}(1/M)$; $\tau_{50\%} = \tau_{\text{swap}}(50\%)$ the instant at which half of the forest has been renewed; τ_{det} the first crossing of λ by the detector state S_t ; and τ_{rec} the return of the error to normal, defined in the paragraph on the four markers below. The two amplitudes below $\Delta e = 0.10$ lie outside the decision domain fixed in Q D.4 and are shaded on every figure that sweeps Δe .

2 The Three Series on a Common Time Axis

No temporal smoothing. e_t is binary. The only averaging applied is the inter-seed mean, taken step by step, with no sliding filter anywhere. A moving average of length L shifts the curve by half its support and destroys the synchronisation with τ^* , which is what this figure exists to display. At the top of the grid the median τ_{ARF} is 28 steps, so a 50-step filter would displace the curve by nearly the whole quantity being measured.

The four markers. τ_{det} is plotted at $\lambda = 8$, the only threshold whose censoring stays in the minority (4.70%, against 61.10% at $\lambda = 25$ and 99.95% at $\lambda = 50$). At the other two thresholds a median delay would rest on a minority of runs, so none is published. τ_{rec} is the return of the mean error below $\hat{p}_0 + \rho \Delta e$ with $\rho = 0.25$, held for 20 consecutive steps; Q C.1 writes it $\tau_{\text{err}}(\rho)$, and the two names denote the same object throughout. The persistence requirement does real work: without it, two amplitudes cross at the very first post-drift step, which is a fluctuation rather than a recovery, and two more cross at step 101, well before the median τ_{ARF} , without the crossing being sustained.

Replacement dynamics. At $t = 100$ the median N_t/M ranges from 0.00 to 0.70 across amplitudes, with an interquartile range of 0.00 to 0.30; at $t = 400$, from 0.10 to 0.90. By the end of the horizon the median runs from 0.60 to 1.00, and the proportion of runs whose forest has been entirely renewed goes from 2% to 94% depending on amplitude.

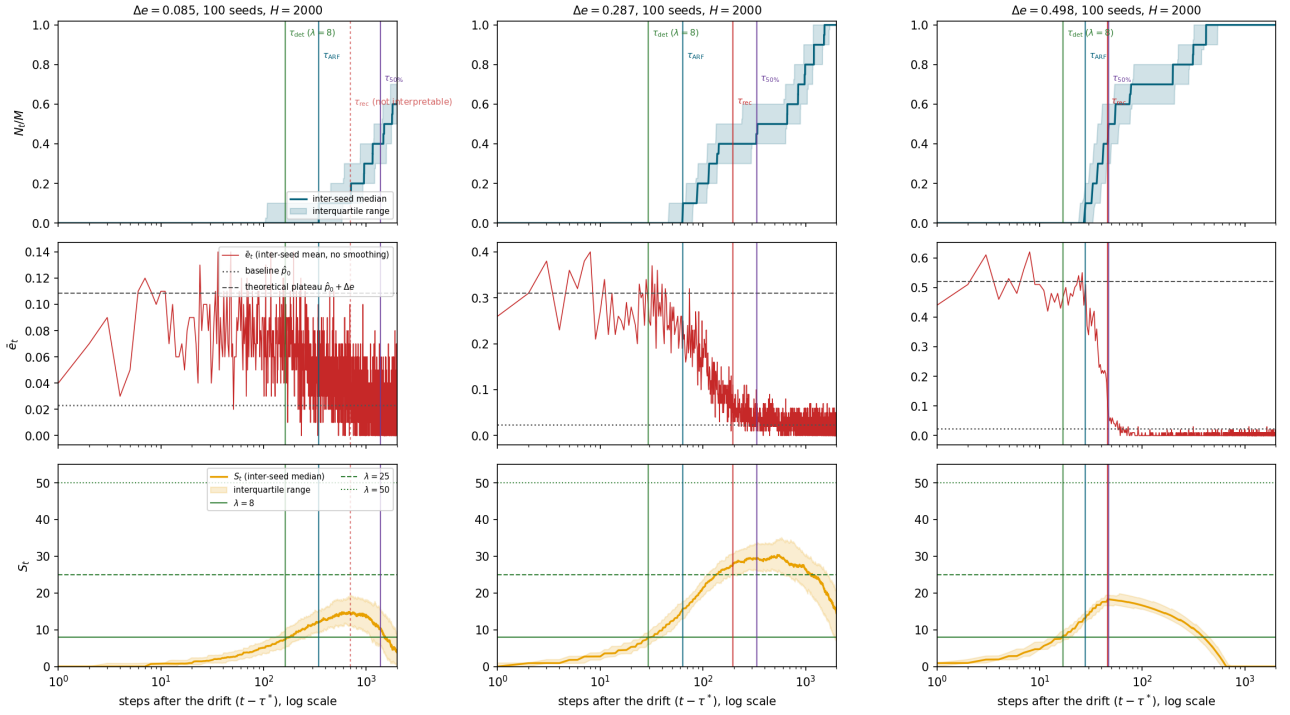


Figure 1: Fraction of replaced trees N_t/M (inter-seed median and interquartile range), global error $\bar{e}_t$, and internal detector state S_t on a common time axis, at three amplitudes spanning the grid, with the four temporal markers τ_{det} (at $\lambda = 8$), τ_{ARF} , $\tau_{50\%}$ and τ_{rec} . The time axis is logarithmic: the four instants span two orders of magnitude across the grid, and on a linear axis the markers of the high-amplitude column collapse into the first few pixels. The τ_{rec} marker is dashed where the test lacks the power to resolve it.

3 The Baseline on Two Windows

The baseline $\hat{p}_0$ governs τ_{det} , $A(H)$ and $S_{\text{max}}(H)$, hence every censoring rate reported in Q D.4. It is therefore reported on both windows the protocol requires: 1 000 pre-drift steps, reproducing R2 as published, and 3 000 steps for the analysis. Since 3 000 pre-drift steps are recorded per run, this is a purely offline recomputation. As a control, the baseline recomputed at 3 000 steps reproduces the campaign value with a maximum deviation of exactly 0.

Window	$\hat{p}_0$ (median)	mean censoring of τ_{det}		
		$\lambda = 8$	$\lambda = 25$	$\lambda = 50$
1 000 steps	0.022000	0.0405	0.6045	0.9990
3 000 steps	0.023000	0.0470	0.6110	0.9995

Table 1: Baseline and its consequences on both estimation windows. The maximum per-run difference between the two baselines is 0.009333.

The window shifts the censoring rates by less than one point and changes no verdict. The direction of the shift is systematic: a baseline estimated over 3 000 steps is slightly higher, so the detector accumulates slightly less evidence and alarms slightly less often.

Against the theoretical baseline. The pre-drift target $y = \mathbf{1}\{x_0 + x_1 > 0\}$ is a deterministic function of the inputs. There is no label noise and the Bayes error is exactly 0, so the measured baseline of 0.023 is entirely approximation error, produced by axis-parallel splits approximating an oblique boundary. The detector monitors departures from a reference level that belongs to the classifier rather than to the problem.

4 What Is Acquired at τ_{ARF}

Two quantities are measured at the instant τ_{ARF} : the fraction of adaptation already acquired, $R(\tau_{\text{ARF}})$, and the fraction of evidence already accumulated by the detector, $G(\tau_{\text{ARF}})$. G is computed run by run, R per amplitude on the mean curve, and τ_{ARF} is the inter-seed median.

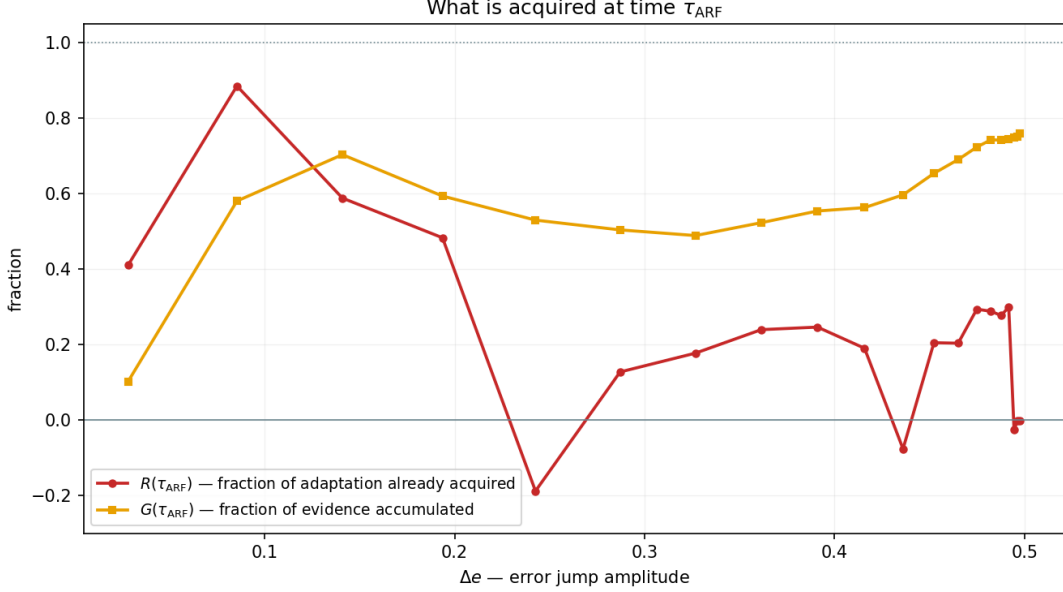


Figure 2: $R(\tau_{\text{ARF}})$ and $G(\tau_{\text{ARF}})$ against Δe . Neither is monotone, and R is negative at five amplitudes.

Answer. G ranges from 0.4889 to 0.7594 over the 18 amplitudes of the decision domain: between half and three quarters of the evidence is already available to the detector when the first tree is replaced. On the full grid it falls to 0.1023, and that restriction is published with the figure.

R supports no such single sentence. It runs from -0.1882 to $+0.8853$ on the grid, from -0.1882 to $+0.5883$ on the decision domain, and it is not monotone in Δe . At $\Delta e = 0.085$ it reaches 0.885: nearly all of the adaptation is acquired when the first tree is replaced, the opposite of what one would expect. At $\Delta e = 0.141$ it is 0.588. The reading “early in the adaptation, late in the race” is not supported by the table and is not used here. What the measurement supports is narrower: at τ_{ARF} , between 49% and 76% of the evidence is accumulated, while the fraction of adaptation acquired varies from negative to 89% with no monotone dependence on amplitude.

A negative fraction of adaptation. R is negative at five amplitudes ($\Delta e = 0.243, 0.436, 0.494, 0.496, 0.498$), reaching -0.1882 . The window used to estimate the post-drift plateau is bounded by $\tau_{\text{ARF}}/2$ precisely to keep it from overlapping the adaptation it measures, and the bound is active at those amplitudes (the window drops to 43, 17, 14, 14, 14 steps respectively). The symptom persists anyway. A negative R means the mean error at τ_{ARF} sits *above* the plateau estimated just after the drift: the error is still rising when the first tree is replaced. Calling the estimation artefact repaired would be false.

The plateau never reaches its theoretical value. The control of the measured plateau against $\hat{p}_0 + \Delta e$ fails on all 20 amplitudes, with a deviation between -0.0271 and -0.0080 , always negative. Two mechanisms account for it, and both point at the baseline.

First, $\hat{p}_0$ is partly a product of the replacement mechanism. On the drift-free arm of the campaign (2000 runs, no change point at all), the forest still performs a median of 14 replacements and renews 9 of its 10 trees over the horizon, for a mean error of 0.0228. Replacement churn under pure noise is the normal regime. A direct comparison against a forest that never replaces anything gave 0.0125, but that figure is not produced by this campaign: it comes from an exploratory measurement of 12 executions (`resultats_R2/JOURNAL.md` § 2 b) whose output was not persisted, and it is quoted as an

order of magnitude. Establishing the permanent cost of the replacement mechanism properly is an open item of that journal.

Second, at large amplitudes the post-drift task is easier than the original one, so the error does not settle at $\hat{p}_0 + \Delta e$ at all. The stylised model of the subject fails at the top of the grid, and that is a measurement rather than a conjecture.

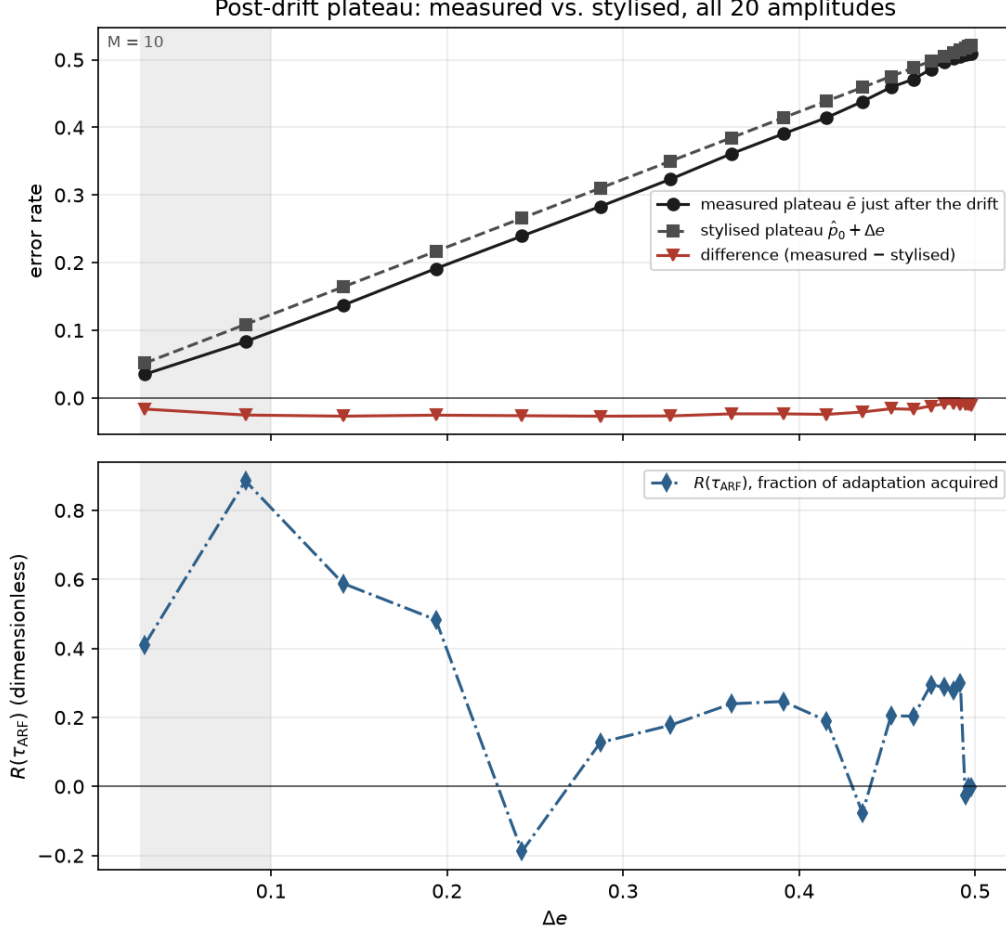


Figure 3: Top: the measured post-drift plateau $\bar{e}$ against the stylised plateau $\hat{p}_0 + \Delta e$, and their difference, on all 20 amplitudes; the difference is negative everywhere, between -0.0271 and -0.0080 . Bottom: $R(\tau_{\text{ARF}})$ with the line at zero, negative at five amplitudes and not monotone. Table QCD_R_et_G_full.

5 An Observation on the Race

Figure 1 makes visible something the aggregate framing of the paradox hides. At $\lambda = 8$, $\tau_{\text{det}} < \tau_{\text{ARF}}$ in 95.44% of the 1906 uncensored runs, or 90.95% of all 2000 runs when censored ones are imputed as losses. Among those 2000 runs, 34 have $\tau_{\text{ARF}} = 0$: a tree is replaced at the change point itself, and the detector loses by construction, since τ_{det} cannot precede the drift. They occur at every amplitude, three at $\Delta e = 0.028$ down to one at the top of the grid, and Q D.2 plots them explicitly. At $\lambda = 25$ the proportion falls to 6.56%, and at $\lambda = 50$ it is 0% on the single uncensored run.

Those grid averages need one qualification, because the rate is far from stationary in Δe . At $\lambda = 8$ the detector wins 0 of the 10 uncensored runs at $\Delta e = 0.028$ (where 90% of runs never alarm at all), 64.6% at $\Delta e = 0.085$ (62 of 96 uncensored runs, Wilson 95% interval 54.6% to 73.4%), and between 88% and 99% everywhere above. The 95.44% figure is dominated by the 18 easy amplitudes out of 20. The blind spot is therefore a property of the pair $(\lambda, \Delta e)$ rather than of λ alone: even at the most sensitive threshold tested, it is fully present at the bottom of the grid. Table 2 gives the race amplitude by amplitude at $\lambda = 8$, and Figure 4 the same for the three thresholds together with the censoring that each cell carries. Both are the material that question A reads as $P(\tau_{\text{ARF}} < \tau_{\text{det}})$ as a function of $(\lambda, \Delta e)$; they are produced here and not interpreted further.

Δe	uncensored / 100	$\tau_{\text{det}} < \tau_{\text{ARF}}$	share of uncensored, % [Wilson 95%]	share of all 100, %
0.028	10	0	not reported (censoring 90%)	0.0
0.085	96	62	64.6 [54.6; 73.4]	62.0
0.141	100	88	88.0 [80.2; 93.0]	88.0
0.194	100	94	94.0 [87.5; 97.2]	94.0
0.243	100	97	97.0 [91.5; 99.0]	97.0
0.287	100	97	97.0 [91.5; 99.0]	97.0
0.327	100	97	97.0 [91.5; 99.0]	97.0
0.361	100	98	98.0 [93.0; 99.4]	98.0
0.391	100	98	98.0 [93.0; 99.4]	98.0
0.416	100	98	98.0 [93.0; 99.4]	98.0
0.436	100	99	99.0 [94.6; 99.8]	99.0
0.452	100	99	99.0 [94.6; 99.8]	99.0
0.465	100	99	99.0 [94.6; 99.8]	99.0
0.475	100	99	99.0 [94.6; 99.8]	99.0
0.482	100	99	99.0 [94.6; 99.8]	99.0
0.488	100	99	99.0 [94.6; 99.8]	99.0
0.492	100	99	99.0 [94.6; 99.8]	99.0
0.494	100	99	99.0 [94.6; 99.8]	99.0
0.496	100	99	99.0 [94.6; 99.8]	99.0
0.498	100	99	99.0 [94.6; 99.8]	99.0
all 20	1 906	1 819	95.44	90.95

Table 2: The race at $\lambda = 8$, amplitude by amplitude. Third column: runs in which the detector alarms strictly before the first tree replacement. Fourth column: that count over the uncensored runs of the cell, with a Wilson 95% interval; it is a statistic on survivors and is not reported where more than half of the cell is censored. Fifth column: the same count over all 100 runs, censored runs counted as detector losses. The last row is the grid total, $1\,819/1\,906 = 95.44\%$ (Wilson 94.4% to 96.3%) and $1\,819/2\,000 = 90.95\%$ (Wilson 89.6% to 92.1%). Those two intervals measure sampling only: both totals pool cells of very unequal difficulty, and it is the spread down the column, not the interval, that carries the qualification. Table `QCD_indicateurs_full`, computed by `chiffres_QD.py`.

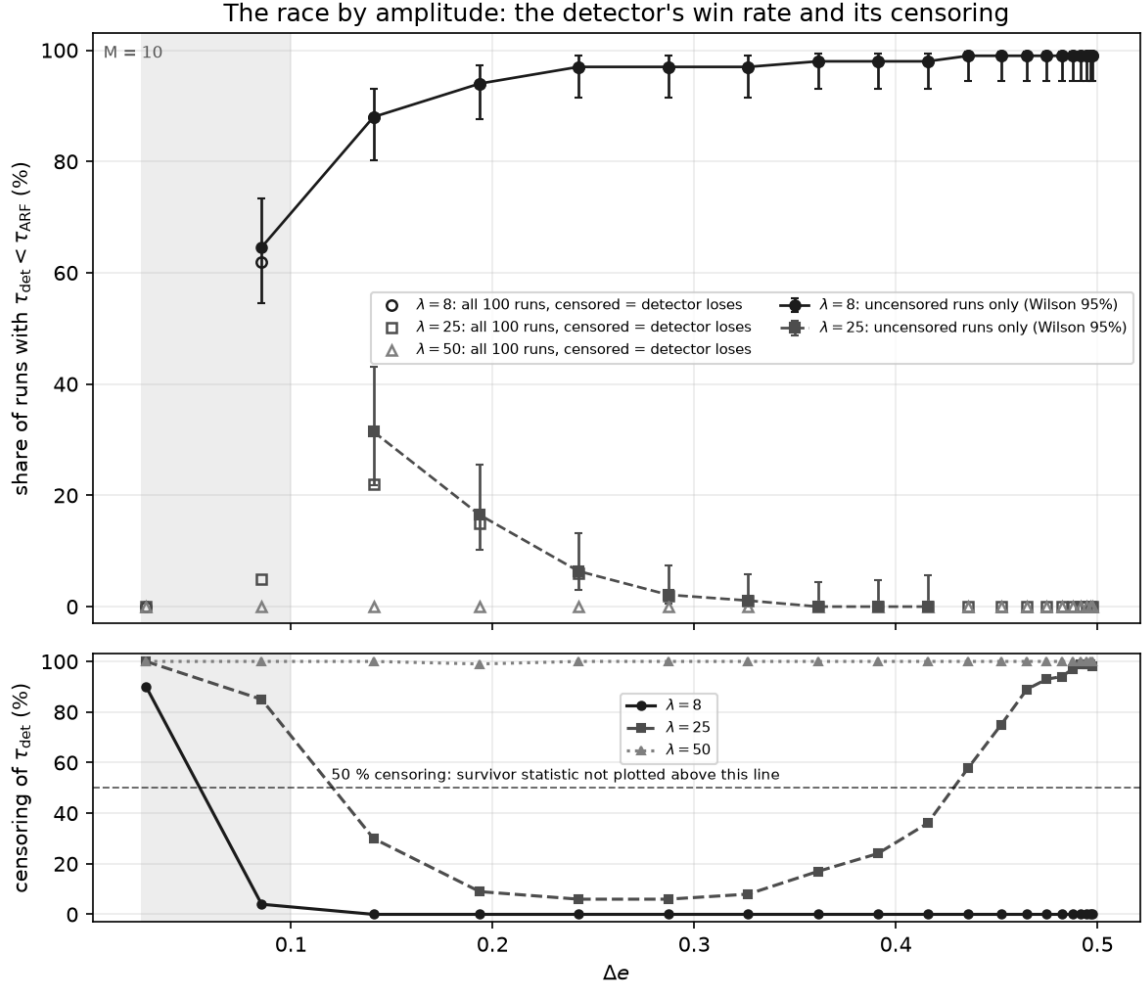


Figure 4: Top: share of runs in which $\tau_{\text{det}} < \tau_{\text{ARF}}$, by amplitude, for $\lambda \in \{8, 25, 50\}$. Filled markers with error bars: uncensored runs only, Wilson 95% interval, drawn only where the censoring of the cell is at most 50%. Hollow markers: all 100 runs, censored runs counted as detector losses, drawn everywhere. Bottom: censoring of τ_{det} in the same cells, with the 50% line. What to see: at $\lambda = 8$ the detector wins almost everywhere except at the bottom of the grid; at $\lambda = 25$ its win rate falls from 31% at $\Delta e = 0.141$ to zero by $\Delta e = 0.36$ while its censoring climbs back to 99% at the top of the grid; at $\lambda = 50$ it never wins. The fractions plotted are those of Table 2, checked for equality inside the script.

6 Not Addressed

The official statement of question D asks for the correlation of τ_{ARF} with the residual error and with the effective time of return to normal. The second is not computed as a per-run correlation anywhere in Q D, and the reason belongs here rather than in silence.

$\tau_{\text{err}}(\rho)$ is defined only on the seed-averaged error curve, one value per amplitude: on a single run the binary sequence is far too noisy for a threshold crossing to signal genuine recovery. The correlations of Q D are computed run by run, so the two objects are not commensurable. It appears in Figure 1 as the τ_{rec} marker, the only use its definition supports, dashed at the amplitudes where the test has no power to resolve it.

“Return of the error to an acceptable level” also fails as an adaptation criterion, for a reason established elsewhere in this work. At $\Delta e = 0.498$ the threshold $\hat{p}_0 + \rho \Delta e$ is crossed at step 46, with or without the persistence requirement, because the post-drift problem is easier than the original one and the error falls below the pre-drift baseline. A criterion that a simplified problem satisfies immediately does not measure adaptation.

Finally, the campaign exists at $M = 10$ only, so no statement in Q D holds as a function of forest size. The instrumented runs at other values of M were produced for a different test and carry no evidence trace.